

BRANCH AND BOUND

Branch & Bound

- Branch and bound is one of the techniques used for problem solving.
- It is similar to the backtracking since it also uses **the state space tree**.
- It is used for solving the optimization problems and **minimization problems**.
- Branch and bound build the state space tree, one can find the optimal solution quickly by pruning a few of the tree branches that do not satisfy the bound.
- Branch and bound can be useful where some other optimization techniques like greedy or dynamic programming fail.
- In the worst case, it may run in exponential time, but careful selection of bounds and branches makes an algorithm run reasonably faster.

BRANCH AND BOUND

- **Introduction:**
- Branch and Bound refers to all state space search methods in which all children of the E- Node are generated before any other live node becomes the E-Node.
- Branch and Bound is the generalization of both graph search strategies, BFS and D-search.
- A BFS like state space search is called as FIFO (First in first out) search as the list of live nodes in a first in first out.
- A DFS-search like state space search is called as LIFO (last in first out) search as the list of Live nodes in a last in first out list.
- **Live node** is a node that has been generated but whose children have not yet been generated.
- **E-node** is a live node whose children are currently being explored. In other words, an E-node is a node currently being expanded.
- **Dead node** is a generated node that is not be expanded or explored any further. All children of a dead node have already been expanded.
- *Here we will use 3 types of search strategies:*
 - 1.FIFO (First In First Out)
 - 2.LIFO (Last In First Out)
 - 3.LC (Least Cost) Search

FIFO Branch and Bound Search:

- For this we will use a data structure called Queue. Initially Queue is empty.

LIFO Branch and Bound Search:

- For this we will use a data structure called Stack.

0/1 Knapsack Problem

Example: Consider the instance $M=15$, $n=4$, $(p_1, p_2, p_3, p_4) = 10, 10, 12, 18$ and $(w_1, w_2, w_3, w_4)=(2, 4, 6, 9)$.

Solution:

- In this problem we will calculate lower bound and upper bound for each node.

Upper bound calculation:

place the first item in the knapsack. Remaining weight of knapsack is $15-2=13$. Place next item w_2 in knapsack and the remaining weight of knapsack is $13-4=9$. Place next item w_3 , in knapsack then the remaining weight of knapsack is $9-6=3$.

No fraction are allowed in calculation of upper bound so w_4 , cannot be placed in knapsack.

Profit= $p_1+p_2+ p_3=10+10+12$ So, **Upper bound=-32**

Lower bound calculation:

To calculate Lower bound we can place w_4 in knapsack since **fractions are allowed** in calculation of lower bound.

$$\text{Lower bound} = 10 + 10 + 12 + (3/9 * 18) = 32 + 6 = 38$$

- Knapsack is maximization problem but branch bound technique is applicable for only minimization problems. In order to convert maximization problem into minimization problem we have to take negative sign for upper bound and lower bound. **GUB=-32**

- Upper bound (U) = -32
- Lower bound (L) = -38

$$\begin{array}{l} U = -32 \\ L = -38 \end{array} \quad \textcircled{1}$$

$(p_1, p_2, p_3, p_4) = 10, 10, 12, 18$ and $(w_1, w_2, w_3, w_4) = (2, 4, 6, 9)$.
 GUB = -32

For node 2, $x_1=1$, means we should place first item in the knapsack.

$U=10+10+12=32$, make it as -32

$L=10+10+12+ (3/9*18) = 32+6=38$,

we make it as -38

Node 2 is expanded

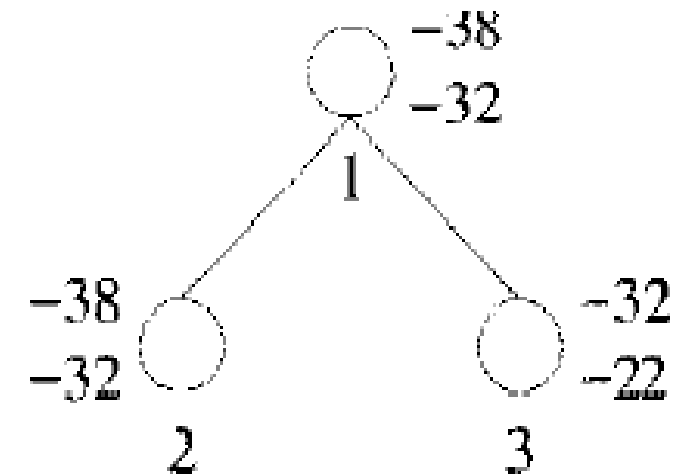
For node 3, $x_1=0$, means we should not place first item in the knapsack.

$U=10+12=22$, make it as -22

$L=10+12+ (5/9*18) = 10+12+10=32$,

we make it as -32

Node 3 is expanded



Upper number = L

Lower number = U

3
2
1

$(p_1, p_2, p_3, p_4) = 10, 10, 12, 18$ and $(w_1, w_2, w_3, w_4) = (2, 4, 6, 9)$.

• For Node 4 : GUB = -32

U = -32

L = -38

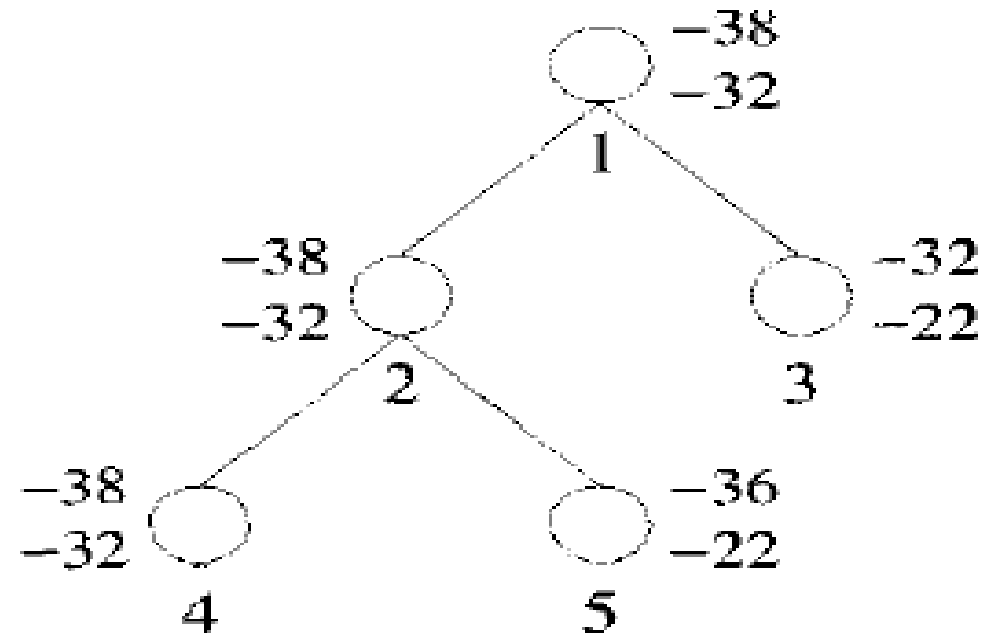
4 are expanded

For Node 5 :

U = -22

L = -36

5 are expanded



Upper number = L
Lower number = U

$(p_1, p_2, p_3, p_4) = 10, 10, 12, 18$ and $(w_1, w_2, w_3, w_4) = (2, 4, 6, 9)$.

For node 7:

GUB = -32

Node 7 is immediately killed as

$L(7) > \text{upper}$.

For node 8:

$U = -32$

$L = -38$

Node 8 is expanded

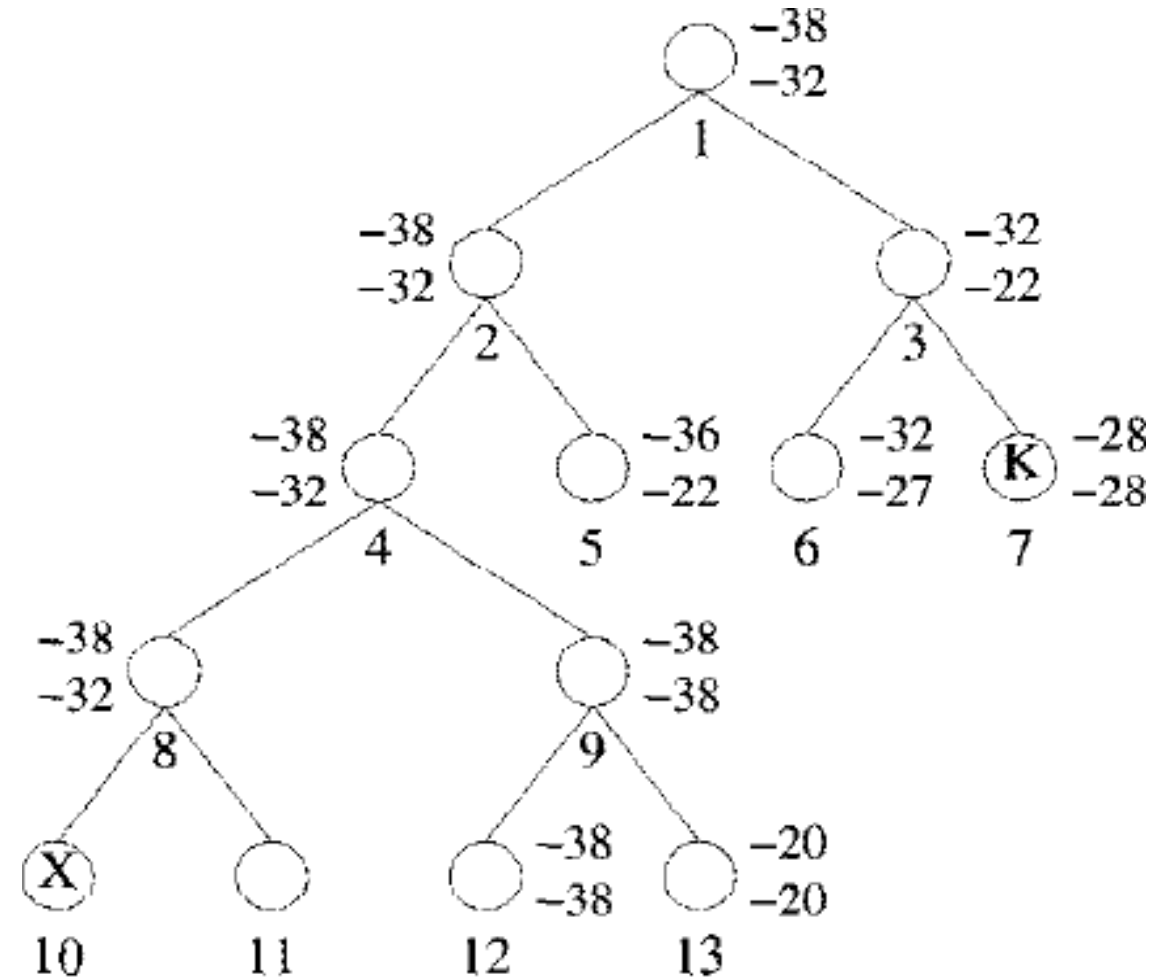
For node 9:

GUB = -38

$U = -38$

$L = -38$

Node 8 is expanded, **GUB is updated**



upper number = L
lower number = U

$(p_1, p_2, p_3, p_4) = 10, 10, 12, 18$ and $(w_1, w_2, w_3, w_4) = (2, 4, 6, 9)$.

For node 10:

GUB = -38

Node 10 is infeasible and so killed

For node 11:

U = -22 L = -32

Node 11 is killed

For node 12:

GUB = -38

U = -38 L = -38

Node 12 is expanded

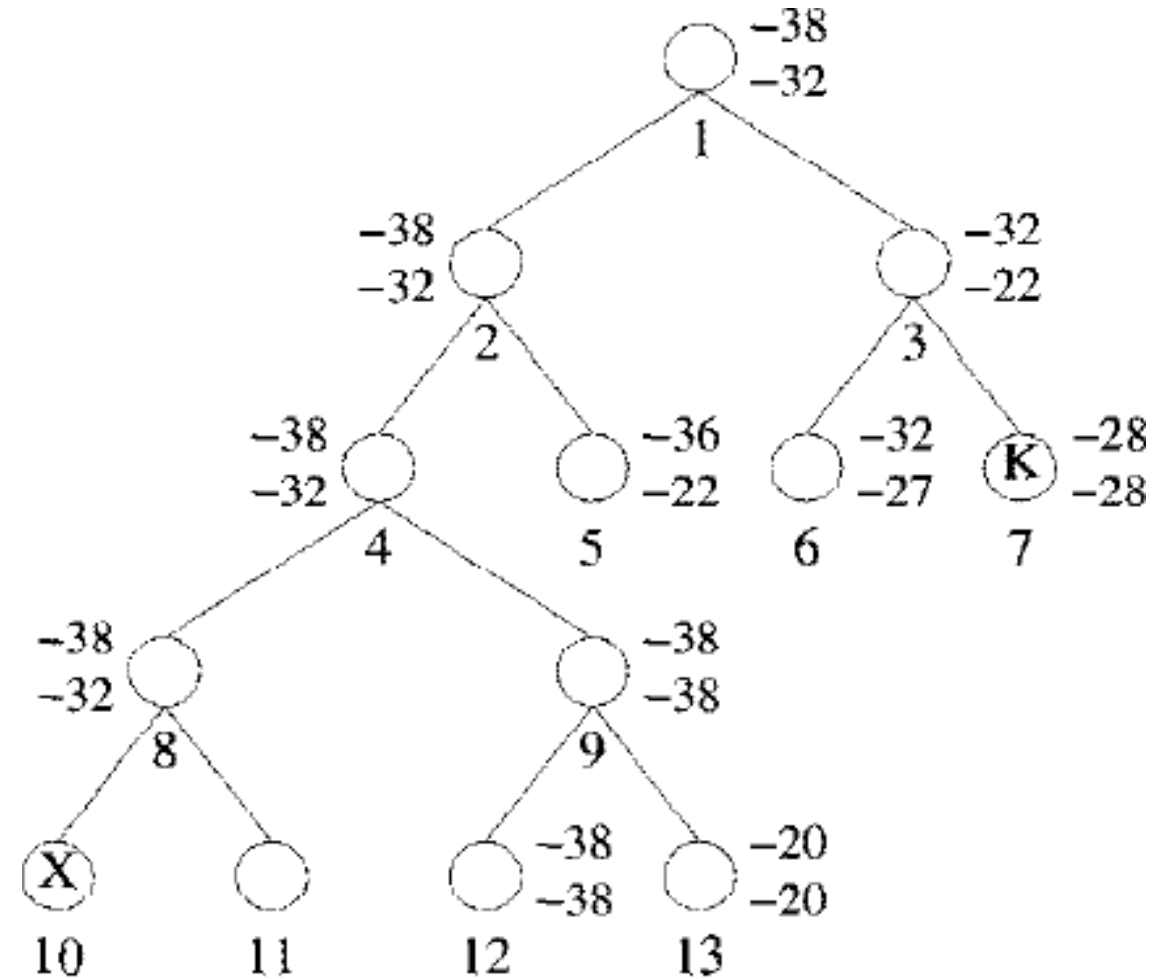
For node 13:

GUB = -38

U = -20 L = -20

Node 13 is killed

Solution is [1,1,0,1] total value = 10 + 10 + 18 = 38



Time and Space Complexity:

The worst case time complexity is given by $O(2^n)$, in cases where the entire tree has to be explored.

Hence its time complexity will be exponential.

0/1 Knapsack problem using LIFO Branch and Bound

LIFO

Day

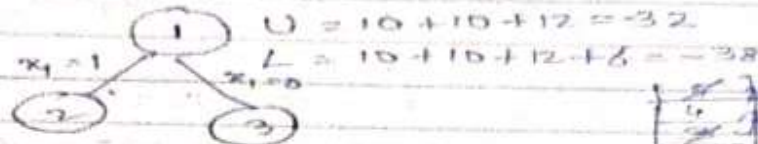
$$M = 15$$

$$n = 4$$

$$P_1, P_2, P_3, P_4 = 10, 10, 12, 18$$

$$W_1, W_2, W_3, W_4 = 2, 4, 6, 7$$

$$GUB = -32$$



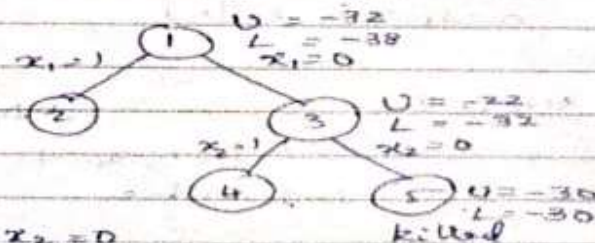
Node 3: $x_1 = 0$

$$U = 10 + 12 = 22$$

$$L = 10 + 12 + 10 = 32$$

Node 3 is expanded.

$$GUB = -32$$



Node 5: $x_1 = 0, x_2 = 0$

$$U = 12 + 18 = 30$$

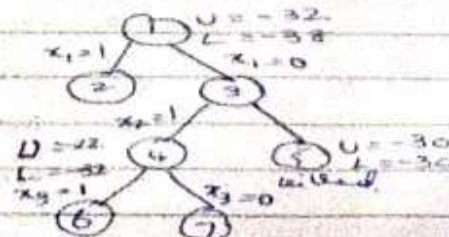
$$L = 12 + 18 = 30$$

Node 4: $x_1 = 0, x_2 = 1$

$$U = 10 + 12 = 22$$

$$L = 10 + 12 + 10 = 32$$

Node 4 is expanded.



$$GUB = -32$$

Node 7: $x_1 = 0, x_2 = 1, x_3 = 0$.

$$U = 10 + 18 = -28$$

$$L = 10 + 18 = -28$$

Node 7 is killed.

7
6
5
4
3
2
1

Node 6: $x_1 = 0, x_2 = 1, x_3 = 1$.

$$U = 10 + 12 = -22$$

$$L = 10 + 12 + 10 = -32$$

Node 6 is expanded.

Node 9: $x_1 = 0, x_2 = 1, x_3 = 1, x_4 = 0$.

$$U = 10 + 12 = -22$$

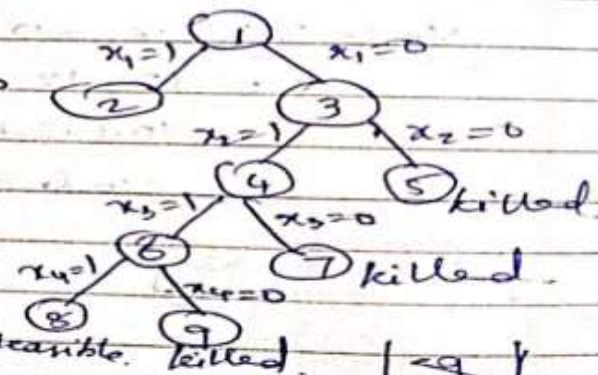
$$L = 10 + 12 = -22$$

Node 9 is killed.

Node 8:

$$x_1 = 0, x_2 = 1, x_3 = 1, x_4 = 1$$

infeasible



9
8
6
2

Node 2: $x_1 = 1$

$$U = 10 + 10 + 12 = -32$$

$$L = 10 + 10 + 12 + 6 = -38$$

Node 2 is expanded

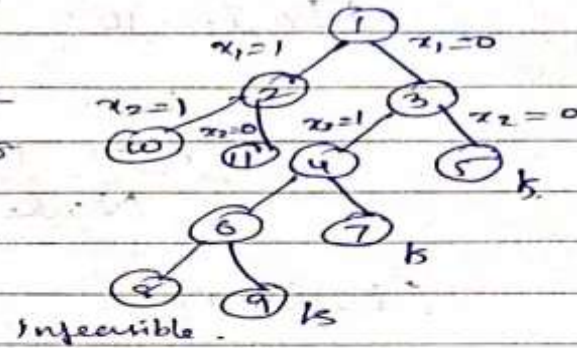
11
10
2

Node 11: $x_1=1, x_2=1$

$$U = 10 + 10 + 12 = -32$$

$$L = 10 + 10 + 12 + 6 = -38$$

Node 11 is expanded.



Node 13: $x_1=1, x_2=0, x_3=0$

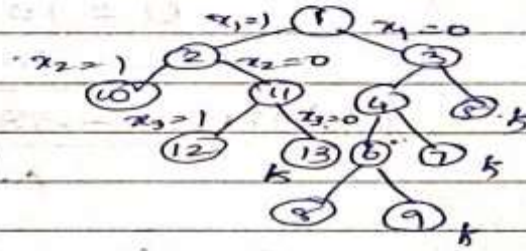
$$U = 10 + 18 = -28$$

$$L = 10 + 18 = -28$$

13 is killed.

13
12
11
10

GUB = -32

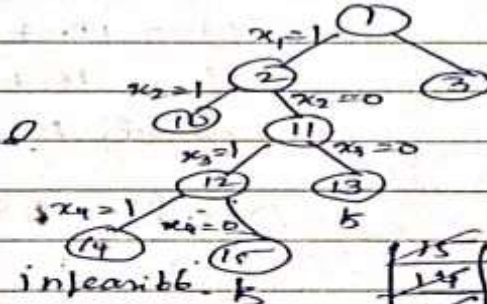


Node 12: $x_1=1, x_2=0, x_3=1$

$$U = 10 + 12 = -22$$

$$L = 10 + 12 + 14 = -36$$

Node 12 is expanded.



Node 15: $x_1=1, x_2=0, x_3=1, x_4=0$

$$U = 10 + 12 = -22$$

$$L = 10 + 12 = -22$$

Node 15 is killed.

15
14
12
10

Node 14: $x_1=1, x_2=0, x_3=1, x_4=1$

$$U = 10 + 12 + 18$$

Node 14 is infeasible.

$$\boxed{GUB = -32}$$

Node 10: $x_1 = 1, x_2 = 1$

$$U = 10 + 10 + 12 = -32$$

$$L = 10 + 10 + 12 + 6 = -38$$

Node 10 is expanded.

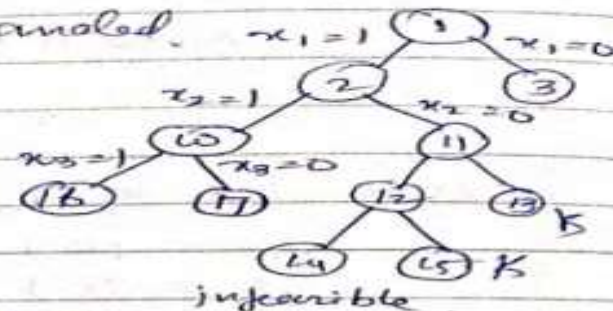
17
16
15

Node 17:

$$x_1 = 1, x_2 = 1, x_3 = 0$$

$$U = 10 + 10 + 18 = -38$$

$$L = 10 + 10 + 18 = -38$$



$$\boxed{GUB = -38}$$

GUB is updated

Node 17 is expanded

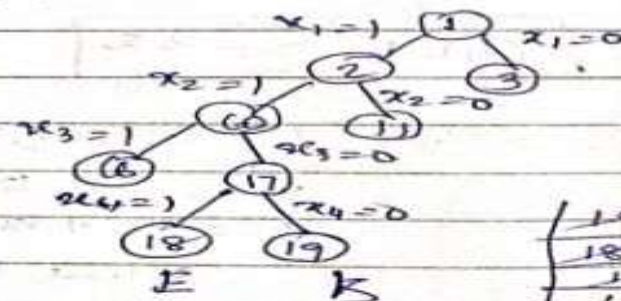
Node 19:

$$x_1 = 1, x_2 = 1, x_3 = 0, x_4 = 0$$

$$U = 10 + 10 = -20$$

$$L = 10 + 10 = -20$$

Node 19 is killed



19
18
17
16

Node 18:

$$x_1 = 1, x_2 = 1, x_3 = 0, x_4 = 1$$

$$U = 10 + 10 + 18 = -38$$

$$L = 10 + 10 + 18 = -38$$

Node 18 is expanded

Node 16: $x_1 = 1, x_2 = 1, x_3 = 1$

$$U = 10 + 10 + 12 = -32$$

$$L = 10 + 10 + 12 + 6 = -38 \text{ is expanded.}$$

For Node 21:

$$x_1 = 1, x_2 = 1, x_3 = 1, x_4 = 0$$

infeasible.

$$U = 10 + 10 + 12 = -32$$

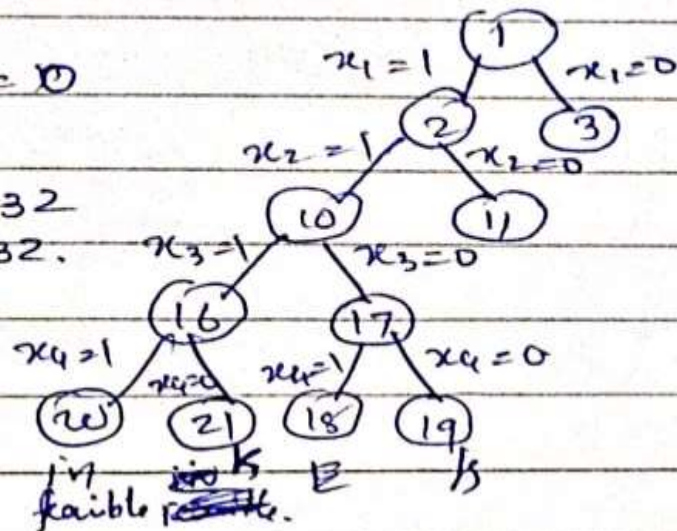
$$L = 10 + 10 + 12 = -32.$$

killed.

For Node 20:

$$x_1 = 1, x_2 = 1, x_3 = 1, x_4 = 1$$

infeasible.



21
20
16

21
20

So solution is $[1, 1, 0, 1]$